



XMBA 201A
Economic Analysis for Business Decisions

Monopoly Pricing

Lecture 4

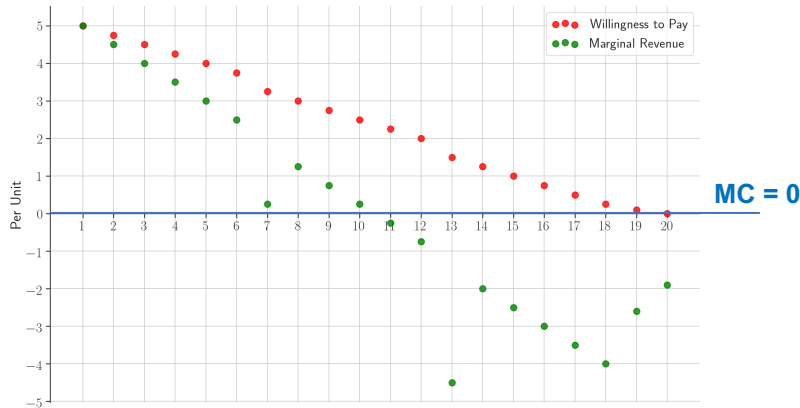
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Recap

- **How much to sell (price)**: every unit for which the income on that unit (marginal revenue) is greater than the marginal cost
- This simple rule is the backbone for monopoly pricing.
- Therefore, MC and demand (MR) determine the optimal price/quantity
- $AC(q)$ is not playing a role for how to price or how much to produce, ***only matters for if to produce at all!***

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With $MC = 0$, profit maximizing price is \$2.5, 10 sold and revenues is \$25.

Taylor Swift: How does this relate?

1. Is it possible that profit-maximizing prices can result in empty seats?
2. What is the effect of Ticketmaster's Verified Fan program on possibility to scalp tickets?

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Taylor Swift

1. $MR = MC (= 0)$ may result in empty seats! The "Reputation" tour did not sell out, yet at the 17 stadiums played on both tours, it grossed 15% more than the 2015 tour.
 - **Price discrimination:** Prices run from \$50 to \$1,500. Empty seats are most likely in the cheapest tiers, where cutting prices would undercut the premium sections (more tomorrow).
 - **Demand uncertainty:** Prices are set before demand is known. High prices impose risk of empty seats in exchange for higher revenue when demand turns out strong.
 - **Complementary revenue:** Attendance also drives sales of merchandise, concessions... so empty seats are less likely than the pure ticket model suggests.
 - **Commitment.** If the seller is expected to cut prices as the show approaches, buyers will wait.
2. Scalping pays when the resale price exceeds face value. Verified Fan works on both the margin and the access:
 - Higher face values imposes risk of unsold inventory by scalpers.
 - Screening on loyalty gives presale access to registered fans rather than bots. Costly signals, such as buying the album, favor buyers who intend to attend.

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The General Construction

- Start with a demand curve: $p(q)$ (easier said than done!)
- Revenue is derived from demand: $R(q) = p(q) \times q$
- $MR(q)$ is derived from $R(q)$ (either using table, or using derivative)
- $MC(q)$ is derived from variable costs
- Optimal production is determined by looking for q^* that satisfies:
 1. $MR(q) > MC(q)$ for $q < q^*$ (or “just to the left” of q^*)
 2. $MR(q) < MC(q)$ for $q > q^*$ (or “just to the right” of q^*)
 3. If there is more than one q^* then choose the one which maximizes profit
 4. At this q^* the profit must be positive, **otherwise shutdown**.

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A Differential Approach (convenient!)

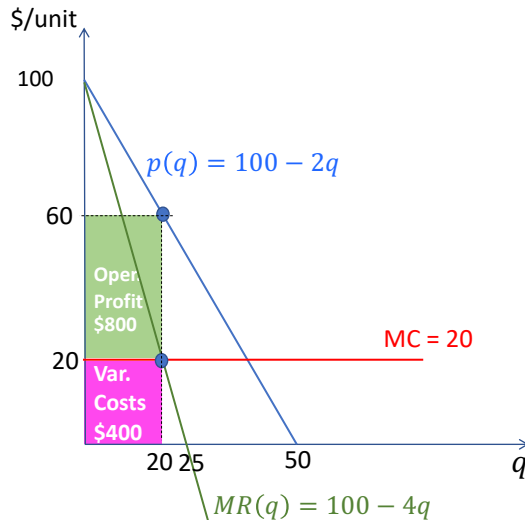
- Assume the demand curve $p(q)$ is differentiable, which means that the derivative $p'(q)$ (or $\frac{dp}{dq}$) is well defined (there are no “jumps” or “kinks”).
- $R(q) = p(q) \times q$ (by definition!);
- $MR(q)$ is the derivative of $R(q)$ at q :

$$MR(q) = R'(q) = p'(q)q + p(q) \text{ (product rule)}$$
- **Example:** Linear Demand, $p(q) = a - bq$
 - $R(q) = aq - bq^2$; $MR(q) = a - 2bq$

Linear demand: MR has same intercept and twice the slope!
 - “well behaved”: $MR(q)$ starts high (at a) and drops at a rate of $2b$, so if MC is not decreasing, MR crosses MC from above at most once!
- **Optimal production:** choose quantity q^* that solves $MR(q^*) = MC(q^*)$
- **Optimal price:** p^* on the demand curve that results in quantity q^*
- Remember to check q^* against shutdown! (Fixed costs)

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A Numerical Example:



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$$\text{Demand: } p(q) = 100 - 2q$$

$$\text{Revenue: } R(q) = p(q) \times q = 100q - 2q^2$$

$$MR(q) = R'(q) = 100 - 4q$$

Linear demand:

- Same intercept
- Twice the slope

$$\text{Total cost: } C(q) = 500 + 20q$$

$$\text{Marginal costs: } MC = 20$$

Optimal quantity:

$$\text{Solve: } MR(q) = MC \text{ for optimal } q$$

$$100 - 4q = 20 \Leftrightarrow q = 20$$

Then take $q = 20$, plug into demand to get the optimal price:

$$p(20) = 100 - 2 \times 20 = 60$$

$$\pi(q) = 60 \times 20 - 500 - 20 \times 20 = 300$$

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Getting Fancy: Elasticity

- Price Elasticity of Demand measures the sensitivity of quantity demanded to changes in price:

$$\epsilon_d = \frac{\% \text{ change in quantity demanded}}{\% \text{ change in price}} = \frac{\frac{\Delta q}{q}}{\frac{\Delta p}{p}} = \frac{\Delta q}{q} \times \frac{p}{\Delta p} = \frac{\Delta q}{\Delta p} \times \frac{p}{q} \cong \frac{dq}{dp} \times \frac{p}{q}$$

- Note: because $\frac{dq}{dp} < 0$, $\epsilon_d < 0$ (some just turn it positive)

- **Neat observation:** recall $R(q) = p(q) \times q$ so,

$$MR = \frac{dp}{dq} \times q + p = \frac{1}{dq/dp} \times q + p = \left(\frac{1}{dq/dp} \times \frac{q}{p} + 1 \right) \times p = \left(\frac{1}{\epsilon_d} + 1 \right) \times p$$

- **Facts:** $MR > 0$ if and only if $\epsilon_d < -1$; $MR < 0$ iff $-1 < \epsilon_d < 0$; and $MR = 0$ iff $\epsilon_d = -1$.

- **Consequence:** You always produce on the elastic part of demand: $\epsilon_d \leq -1$!

- When $MR < 0$ you should cut back on quantity!

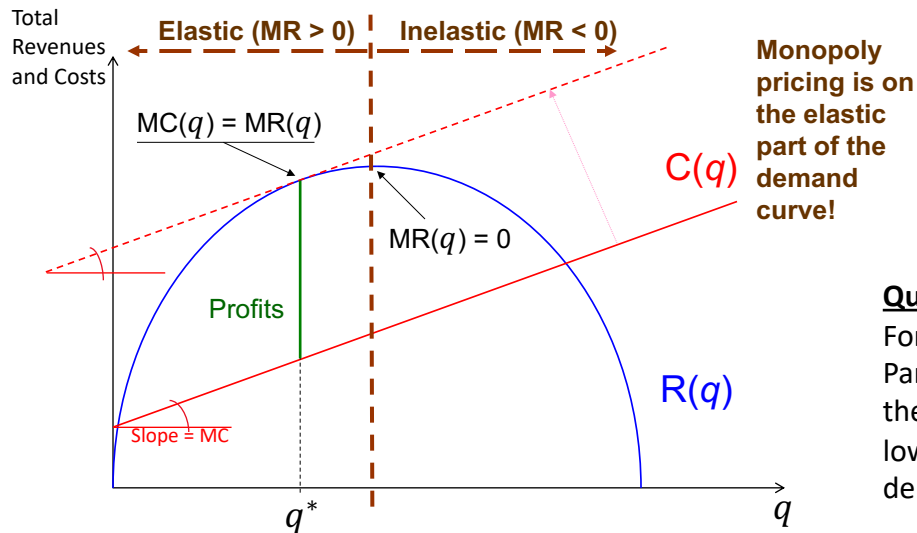
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The Graphical Exposition



Question:

For which products did Parker Hannifin raise their prices: those with low or high elasticities of demand?

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The Learner Rule

- 3 slides back: $R(q) = p(q) \times q$ so,

$$MR = \frac{dp}{dq} \times q + p = \frac{1}{dq/dp} \times q + p = \left(\frac{1}{dq/dp} \times \frac{q}{p} + 1 \right) \times p = \left(\frac{1}{\varepsilon_d} + 1 \right) \times p$$

- Using this equality, the optimality of $MR = MC$ implies that at the optimal price is:

$$\left(\frac{1}{\varepsilon_d} + 1 \right) \times p = MC \Rightarrow p = \frac{MC}{1 + 1/\varepsilon_d}$$

- This is the **Lerner Rule**: You always produce on the elastic part of demand: $\varepsilon_d \leq -1$!

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Seeking Perfect Prices, CEO Tears Up the Rules

1. The article is clear on the pricing scheme that Parker has abandoned, but less clear on what has replaced it. What factors now affect how Parker prices its parts?
2. The article suggests that the standard markup rule (35% in Parker's case) might make sense for a firm selling commodities, but not for a firm selling a portfolio of niche products. If you were selling a commodity product, would a standard markup rule be a good strategy?
3. Firms and managers are often concerned about market share. How is the new pricing scheme likely to change the market shares of Parker's products?

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Seeking Perfect Prices, CEO Tears Up the Rules

1. What now drives Parker's prices?
Customers' willingness to pay, which reflects demand elasticity (**Lerner rule**). The key factors are competition, product superiority, customer preferences (e.g., the 15% filter price cut), and service such as fast delivery.
2. Is a standard markup good for a commodity?
No. A commodity seller is close to a price taker with highly elastic demand, so the optimal markup is near zero. A fixed markup either prices the firm out of the market or leaves money on the table. Match the market price!
3. Effect on market share?
Share falls modestly where prices rose and rises where they fell. Losses are small because niche demand is inelastic: only 3 of 50 repriced parts lost the customer. Parker trades some share for higher margins and profits.

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Some Notes on Demand

- Demand is fixed “other things equal”
- Demand may change. Why?
 - tastes
 - Population growth/decline
 - Income changes (normal (+), inferior (-))
 - Introduction of other substitute (-) or complement (+) goods
- Any such change “shifts” the demand curve
- As a result, optimal price will often change
- **Caveat:** a *change in demand* (shift of curve) is different from a *change in quantity demanded* (move along curve)

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Demand Aggregation

- Assume two sources for chocolate demand
 - Many poor undergrads, few rich MBAs
 - Undergrads: $q_u = 5,000 - 2,500p$ ($\rightarrow p(q) = 2 - q/2,500$)
 - MBAs: $q_m = 1,000 - 200p$ ($\rightarrow p(q) = 5 - q/200$)
- Costs of production:
 - $C(q) = 100 + q$
- What's the demand curve that the chocolate firm faces?
 - Demand is quantity as a function of price
 - Right way to think: for a given p , how much will I sell?
- Aggregating demands:
 - Need to add up the two quantities for any given price to get $Q(p)$
 - Can then get inverse demand $p(Q)$ to derive MR

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Demand Aggregation and Pricing

- Aggregating demand is tricky:

- If $p > 2$ then *only the MBAs buy!*

$$q_u = 5,000 - 2,500p$$

$$q_m = 1,000 - 200p$$

$$Q = q_u + q_m = \begin{cases} 1,000 - 200p & \text{if } 5 \geq p \geq 2 \text{ (MBAs only)} \\ 6,000 - 2,700p & \text{if } p \leq 2 \text{ (MBAs and undergrads)} \end{cases}$$

$$(5,000 - 2,500p) + (1,000 - 200p)$$

- Now we have to figure out $p(Q)$:

- For $5 \geq p > 2$, use MBA demand to obtain correct range $600 \geq Q > 0$
- For $2 \geq p > 0$, use sum of demands to obtain correct range $6,000 \geq Q > 600$

$$p(Q) = \begin{cases} 5 - \frac{Q}{200} & \text{if } 600 \geq Q \geq 0 \\ 2.22 - \frac{Q}{2,700} & \text{if } 6,000 \geq Q \geq 600 \end{cases}$$

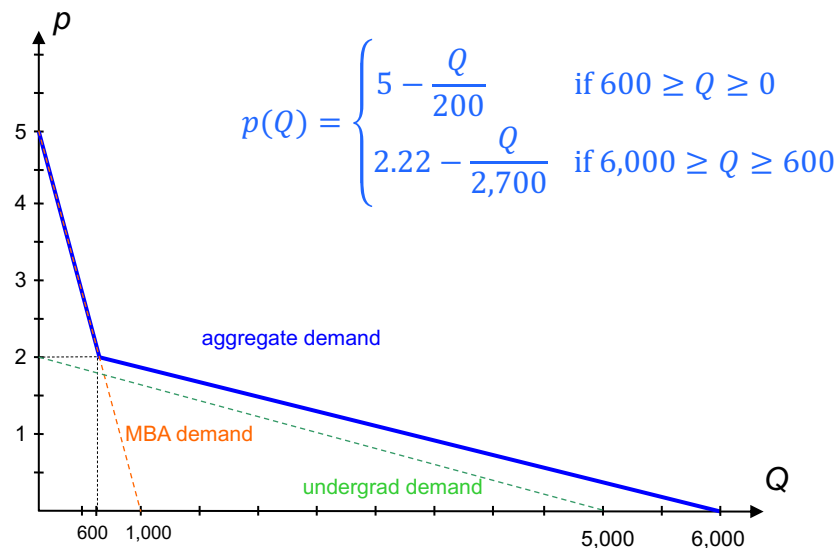
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Aggregation: Graphical Representation



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Demand Aggregation : Deriving MR

- Is deriving MR tricky?
- **Not really!** Just use the right revenue function

$$R(Q) = p(Q)Q = \begin{cases} 5Q - \frac{Q^2}{200} & \text{if } 600 \geq Q \geq 0 \\ 2.22Q - \frac{Q^2}{2,700} & \text{if } 6,000 \geq Q \geq 600 \end{cases}$$

- Now we can derive $MR(Q)$ from $R(Q)$:

$$MR(Q) = \begin{cases} 5 - \frac{Q}{100} & \text{if } 600 \geq Q \geq 0 \\ 2.22 - \frac{Q}{1,350} & \text{if } 6,000 \geq Q \geq 600 \end{cases}$$

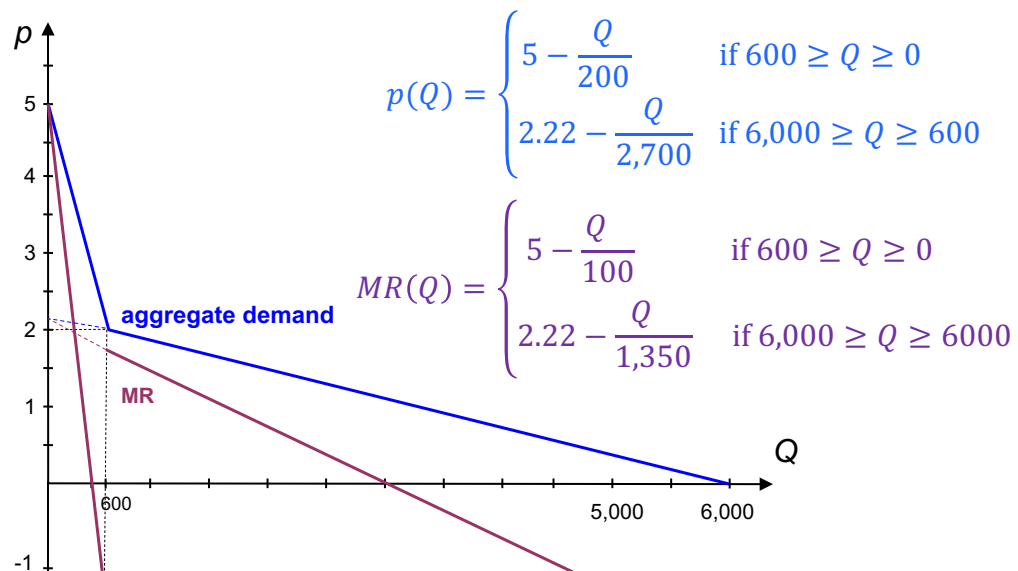
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Aggregation: Graphical Representation



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Aggregation: Optimal Pricing

- Now that we have $MR(Q)$, we only need $MC(Q)$
- From Costs: $C(Q) = 100 + Q$
 $\rightarrow MC(Q) = 1$
- Careful:
 - $MR(Q)$ has two regions
 - $MC(Q)$ may cross $MR(Q)$ twice!
- To see if they cross once or twice need to:
 - Either solve algebra and check that p and Q are consistent
 - Or draw a VERY careful graph

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Optimal Pricing: Algebra

- Recall that:

$$MR(Q) = \begin{cases} 5 - \frac{Q}{100} & \text{if } 600 \geq Q \geq 0 \\ 2.22 - \frac{Q}{1,350} & \text{if } 6,000 \geq Q \geq 600 \end{cases}$$
- $MC(Q) = MR(Q)$ In first region ($600 \geq Q \geq 0$):

$$1 = 5 - Q/100 \rightarrow Q = 400, p = 5 - Q/200 = 3$$

$$\pi = R(Q) - C(Q) = pQ - 100 - Q = \$700$$
- $MC(Q) = MR(Q)$ In second region ($6000 \geq Q \geq 600$):

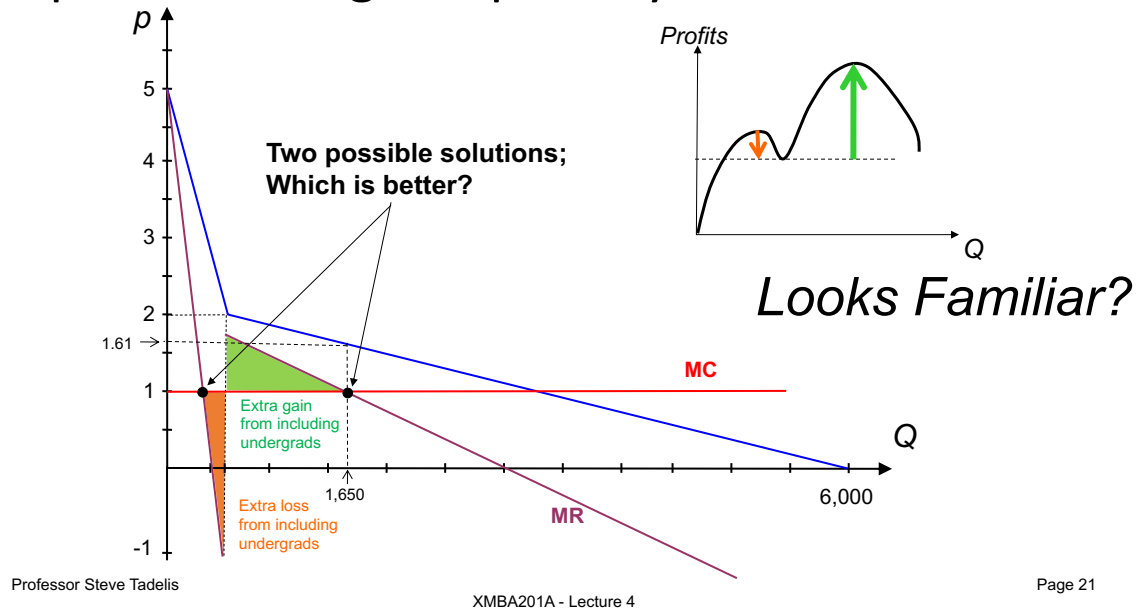
$$1 = 2.22 - Q/1350 \rightarrow Q = 1,650, p = 2.22 - Q/2700 = 1.611$$

$$\pi = R(Q) - C(Q) = pQ - 100 - Q = \$908.33$$
- **Conclusion:** price at \$1.611 and sell to both populations!



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Optimal Pricing: Graphically



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Why discounts?



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Group (3rd Degree) Price Discrimination

- Now distinguish between undergrads and MBAs using their student ID (or their drivers license with mild assumptions on age.)
- This allows us to separate the demands and offer two different prices for two different markets:
 - Undergrads: $q_u = 5,000 - 2,500p_u$ [or $MR_u(q_u) = 2 - q/1,250$]
 - MBAs: $q_m = 1,000 - 200p_m$ [or $MR_m(q_m) = 5 - q/100$]
- What will determine the prices in each market?
 - **Profit maximization!**
- How do we do it?

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Group (3rd Degree) Price Discrimination

- Because MC is constant then it is easy: **two independent equations:**
 $MR_u(q_u) = MC$ and $MR_m(q_m) = MC$
- From $MR_u(q_u) = MC \Rightarrow 2 - q_u/1,250 = 1 \Rightarrow q_u = 1,250$
- From $MR_m(q_m) = MC \Rightarrow 5 - q_m/100 = 1 \Rightarrow q_m = 400$ (not surprising...)
- Plug q's into demand functions to get prices: $p_u = 1.5$; $p_m = 3$
- Profits:

$$\Pi = (1.5 \times 1,250 + 3 \times 400) - 100 - 1 \times (1,250 + 400) = \$1,325$$
- Increased profits by \$416.67! (compared to single price for all students that yielded profits of \$908.33)

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Group (3rd Degree) Price Discrimination

- What happens if MC is not constant? [e.g., $C(Q) = 100 + Q + 0.1Q^2$]
- Now we have many q 's:
 - $MR(q_u)$; $MR(q_m)$; and... **$MC(q_u + q_m)$**
- Some guiding questions – If we are at an optimal point,
 - Can we have $MR(q_u) \neq MC(q_u + q_m)$?
 - Can we have $MR(q_u) \neq MR(q_m)$?
- Solution: $MR(q_u) = MR(q_m) = MC(q_u + q_m)$
 - Two equations with two unknowns (from profit maximization)
 - Same intuition, just a bit more algebra...

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Summary

- Optimal quantity: solve $MR(q) = MC(q)$
- Optimal Price: plug optimal quantity into the demand function $p(q)$
- MC and demand determine optimal quantity; FC and AC only relevant for shut down and NOT optimal pricing!
- Never price on the inelastic part of the demand curve (beware short-vs-long run decisions, e.g., customer loyalty and attrition)
- Aggregating demand:
 - add up $q(p)$ quantities across all relevant markets (notice relevant price regions!)
 - Invert demand to $p(q)$ with relevant quantity regions
 - Derive $MR(q)$ from $p(q)$ with relevant quantity regions
 - Equate $MR(q^*) = MC(q^*)$ then get optimal price from demand $p^*(q^*)$

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